Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738064291165-Chapter 6 - unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

STORY TELLING - HUGH'S GREENHOUSE

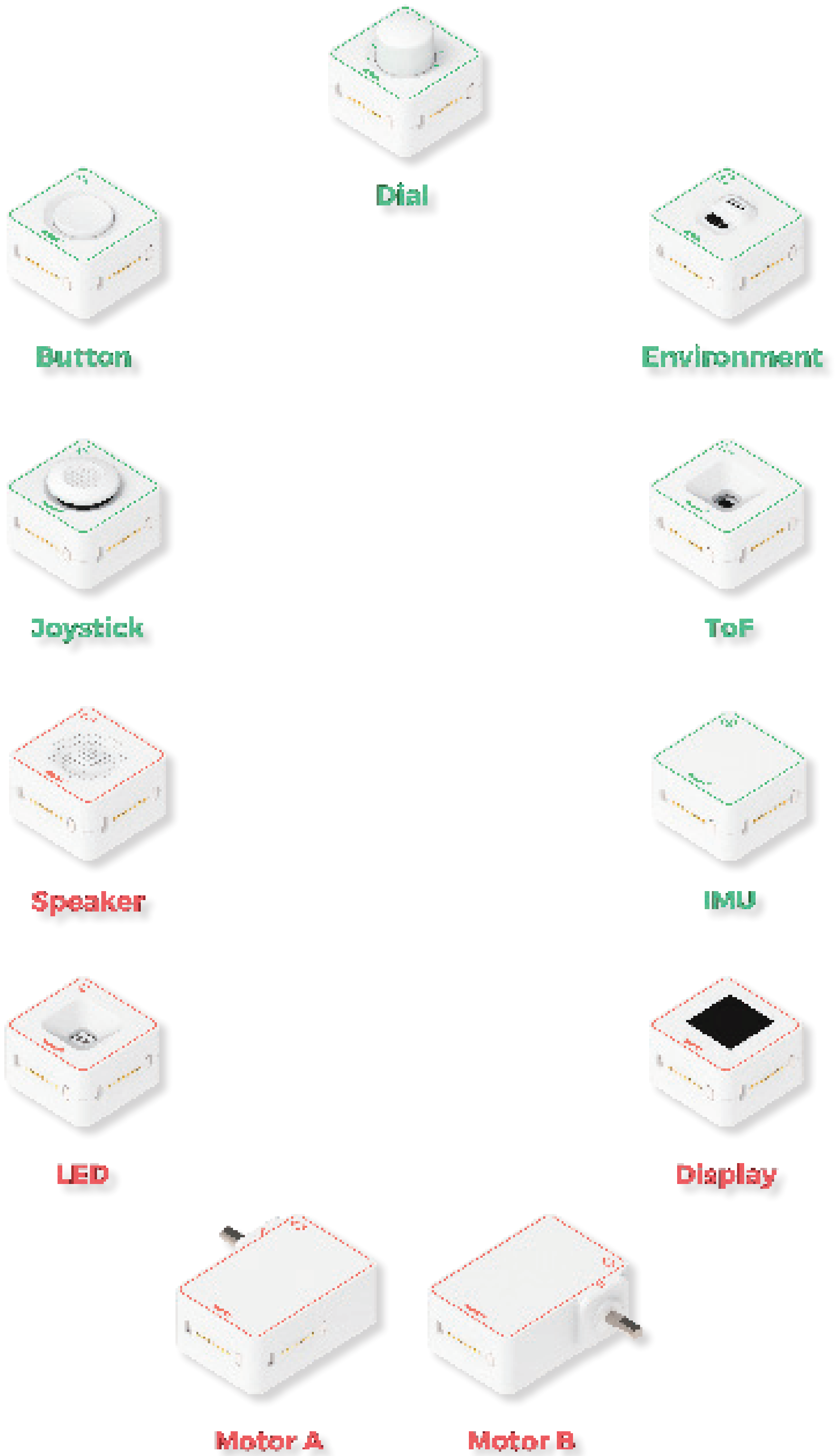
In a small village, Hugh, an innovative farmer, dreamed of a smarter way to tend to his crops. He imagined a device that would automatically adjust the temperature around his plants, cooling them with a soothing blue light on hot days and warming them with a comforting red glow during cold spells.

Hugh's solution came to life with temperature-monitoring sensors and spinning motors assembled with fan wings and a color indicator. When temperatures soared above 25 degrees Celsius, the module glowed blue, activating fans to cool the plants. Conversely, when the temperature dipped below 15 degrees Celsius, it turned red, signaling heaters to warm the greenhouse!

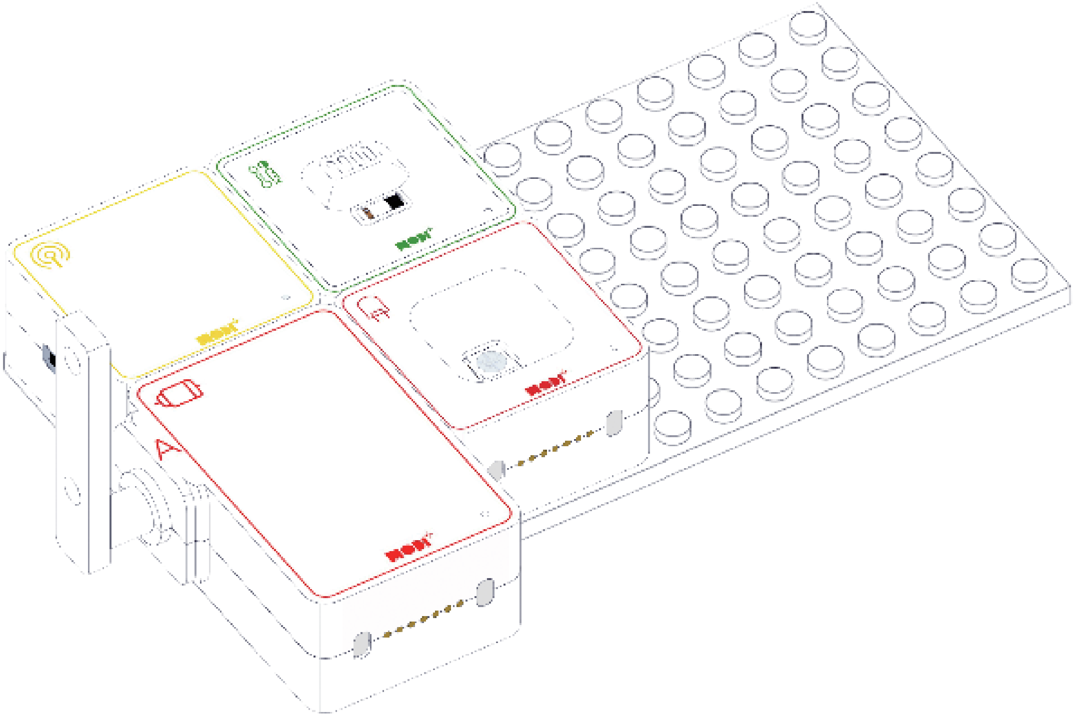
Thanks to this ingenious device, Hugh's farm flourished. "We can build this smart device with just a few modules from the MODI Master Kit!" Hugh realized, eager to share his discovery with fellow farmers.

Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?

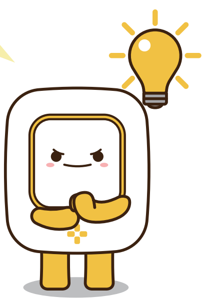


HOW DOES THE CREATION LOOK LIKE?



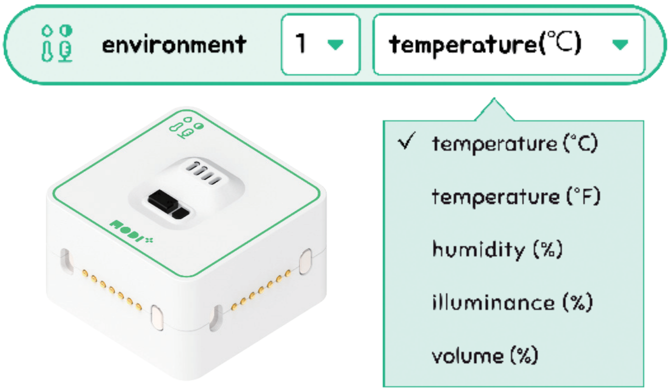
TIPS

Using the MODI Master kit, farmers can ensure plants get enough light and adjust their position or indoor lighting as needed.



Module Functionality

Let's check the details of the input module below for this project.



ENVIRONMENT

- Temperature: Measures how hot or cold thes-urroundings are.
- Humidity: Checks the amount of moisture in the air.
- Illuminance: Gauges the brightness of the light around it.
- Volume: Detects the level of sound in the envi-ronment.

The Environment Module is a smart tool for farmers to make sure their plants are healthy and happy. It checks on three key things plants need:

1. TEMPERATURE MEASUREMENT

Just like we feel too hot or too cold, plants do too! This module keeps an eye on how warm or cool it is, so plants are always comfy.

2. HUMIDITY MEASUREMENT

Plants need just the right amount of water in the air around them. This part of the module measures the air's wetness to help keep plants from getting too thirsty or too soggy.

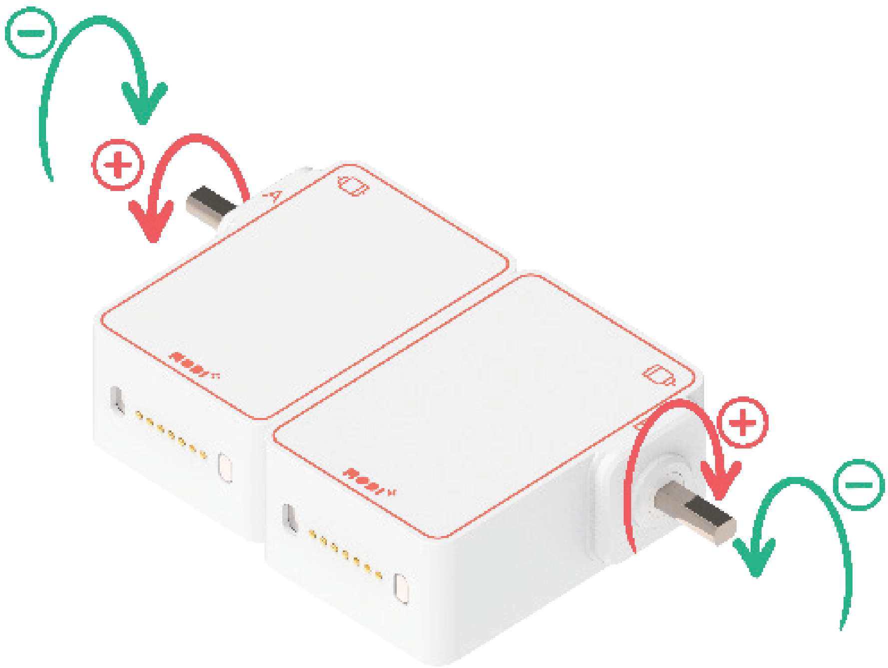
3. LIGHT MEASUREMENT

Plants need light to grow, and this gadget checks if they're getting enough sunshine, making sure they have the energy to grow big and strong.

4. NOISE MEASUREMENT

While plants don't have ears, they can be affected by the sounds around them.
This feature checks how loud or quiet it is, ensuring the plants have a peaceful environment to grow in.

This tine, let's check the details of the output module below for this project



1. Speed	2 . Angle
- Maximum Speed: 198 RPM - Range: -100% to 100% (Negative (%) = Reverse spin) - Direction: + is clockwise, - is counter-clockwise when viewed from axes side	- Range: 0° to 360° - Precision: Exact positioning

TIPS

1) **Direction:** Positive speed values spin both motors opposite to each other, as seen in the image above. .

2) **RPM:** Revolutions per minute; how fast the motor spins.

To understand the motors better, let's try different settings through Code Editor!